

Explain the difference between the terms security, privacy and integrity of data

9608_s21_qp_12_sc_8.a

QUESTION

- 8 (a) Complete the following table by identifying the **most appropriate** term for each description. Each term must be different.

Description	Term
Ensures data is accurate and up to date	
Prevents accidental or malicious data loss	
Prevents unauthorised access to data	

[3]

MARK SCHEME

8(a)	1 mark per term	3								
	<table><tr><th>Description</th><th>Term</th></tr><tr><td>Ensures the data is accurate and up to date</td><td>Data integrity</td></tr><tr><td>Prevents accidental or malicious data loss</td><td>Data security</td></tr><tr><td>Prevents unauthorised access to data</td><td>Data privacy</td></tr></table>	Description	Term	Ensures the data is accurate and up to date	Data integrity	Prevents accidental or malicious data loss	Data security	Prevents unauthorised access to data	Data privacy	
Description	Term									
Ensures the data is accurate and up to date	Data integrity									
Prevents accidental or malicious data loss	Data security									
Prevents unauthorised access to data	Data privacy									

9608_s19_qp_13_sc_2.a

QUESTION

- 2 Frankie is a software developer. He is developing a program to manage customer records for a client with an online retail business. He must ensure that data stored about each customer are both secure and private.

- (a) State the difference between security and privacy.

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..... [2]

MARK SCHEME

2(a)	1 mark per bullet point Security: keep data safe from accidental/malicious damage/loss Privacy: keep data confidential // only seen by authorised personnel	2
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9608_s19_qp_12_sc_5.b.i

QUESTION

(b) Moheem uses a Database Management System (DBMS) to ensure the security and integrity of the data.

(i) Explain the difference between security and integrity.

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..... [2]

MARK SCHEME

5(b)(i)	1 mark per bullet point - Security ensures that data is safe from unauthorised access // safe from loss - Integrity ensures that data is accurate / consistent / up to date	2
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9608_s18_qp_13_sc_2.b

QUESTION

(b) The database manager is concerned about data integrity.

State what is meant by **data integrity**. Give an example of how the manager can ensure data integrity in the PURPLEGAME database.

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..... [2]

MARK SCHEME

2(b)	1 mark for description Ensure data is consistent / accurate // keep data consistent / accurate 1 mark for example from: e.g. - Validation rules - Referential integrity - Verification - Input masks - Setting data types - Removing redundant data - Backup data - Access controls - Audit trail	2
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9608_w17_qp_12_sc_3.a.i

QUESTION

3 A Local Area Network is used by school staff who access data stored in a Database Management System (DBMS).

(a) (i) Explain the difference between security and privacy of data.

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.....[3]

MARK SCHEME

3(a)(i)	1 Mark per bullet, max 3 Security is keeping the data safe From accidental / malicious damage /loss By example of need for security Privacy is the need to restrict access to personal data To avoid it being seen by unauthorised people By example of need for privacy	3
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9608_w17_qp_12_sc_3.a.ii

QUESTION

(ii) Give an example for this application where privacy of data is a key concern.

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.....[1]

MARK SCHEME

3(a)(ii)	1 Mark for a suitable example For example: Personal data of students / staff	1
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9608_s16_qp_13_sc_7.a

QUESTION

7 A bank holds personal data about its customers and their financial data.

(a) Describe the difference between security and integrity of data.

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.....[4]

MARK SCHEME

7	(a) Four from:	[4]
• Security is keeping the data safe. • Integrity is making sure that the data is correct / valid. • Security is the prevention of data loss. • Integrity ensures that the data received is the same as the data sent / data copied is the same as the original. • Example of ensuring security, e.g. usernames and passwords, firewalls etc... • Example of ensuring integrity, e.g. parity checks, double entry etc...		

QUESTION

(b) Describe **two** differences between data integrity and data security.

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..... [2]

MARK SCHEME

(b) **one** mark for security, **one** mark for integrity:

- integrity deals with validity of data/freedom from errors/data is reasonable
- security deals with protection of data
- security protects data from illegal access/loss
- integrity deals with making sure data is not corrupted after, for example, being transmitted

[2]

Show appreciation of the need for both the security of data and the security of the computer system

9608_s19_qp_11_sc_4.b

QUESTION

- (b) The software development company uses data backup and disk-mirroring to keep their data secure.

Explain how data backup and disk-mirroring allow the company to recover from data loss.

Data backup

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Disk-mirroring

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.....

[4]

MARK SCHEME

4(b)	1 mark per bullet point to max 2 × 2	4
	Data backup A copy of data will have been made and stored elsewhere. If the original is lost, the backup can be used to restore the data.	
	Disk-mirroring The data is stored on two disks simultaneously. If the first disk drive fails, the data is accessed from the second disk.	

QUESTION

- (c) Employees using the new computers receive training. At the end of the training, each employee completes a series of questions.

Three answers given by an employee are shown below.

Explain why each answer is incorrect.

- (i) *"Encryption prevents hackers breaking into the company's computers."*

.....

[2]

MARK SCHEME

- (c) (i) • Hackers can still access the data (and corrupt it, change it or delete it)
 • Encryption simply makes data incomprehensible (without decryption key / algorithm)
 [2]

QUESTION

- (b) All of the uses in **part (a)** involve the input or output of data.

- (i) Describe **two** methods of preventing accidental loss of data.

1

 2
[2]

MARK SCHEME

- (b) (i) Any **two** from:
- frequent (or equivalent) backup EITHER to secondary media/to 3rd party server/cloud/removable devices/continuous backup OR stored remotely
 - disk-mirroring strategy/RAID
 - UPS (uninterruptable power supply)/backup generator

[2]

Describe security measures designed to protect computer systems, ranging from the stand-alone PC to a network of computers

9608_s21_qp_12_sc_8.b

QUESTION

(b) Describe what is meant by a **digital signature**.

[2]

MARK SCHEME

8(b)	1 mark per bullet point to max 2 - Mathematical algorithm // encrypted data - Attached to electronically transmitted document ...to verify its content/that it comes from a trusted source/	2
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9608_s21_qp_11_sc_2.d

QUESTION

(d) Zak's company holds details about clients in a database.

Give **three** security measures that Zak can implement to make sure that only authorised employees can access the data.

1

2

3

[3]

MARK SCHEME

2(d)	1 mark per bullet point to max 3 • Password • Biometrics • Access rights • Swipe cards • Physical access measures, e.g. security guards • Implement a firewall/proxy to monitor remote access requests • Two-step authentication	3
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9608_s21_qp_11_sc_9.b

QUESTION

- (b) Complete the table by identifying the **most appropriate** term for each description. Each term must be different.

Description	Term
Receives data packets from a network and forwards them onto a similar network	
Manages access to a centralised resource	
Joins networks that use different sets of rules to transmit data	
Monitors and controls incoming and outgoing network traffic based on set criteria	

[4]

MARK SCHEME

9(b)

1 mark for each correct term

4

Description	Term
Receives data packets from a network and forwards them onto a similar network	Router
Manages access to a centralised resource	Server
Joins networks that use different sets of rules to transmit data	Gateway
Monitors and controls incoming and outgoing network traffic based on set criteria	Firewall

QUESTION

5 Wei is developing a program.

(a) He wants to make sure the source code is secure on his laptop.

Explain how encrypting the source code can keep it secure.

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.....

.....

.....

.....

..... [3]

MARK SCHEME

5(a)	1 mark per bullet point to max 3 • Encryption scrambles the source code (so it is meaningless) • ... using an encryption key / algorithm • If the file is accessed without authorisation it will be meaningless • It requires a decryption key / algorithm to unscramble	3
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QUESTION

(d) The company needs to keep the data on its servers secure from online threats.

(i) Describe how a firewall will help to protect the data on the servers from online threats.

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..... [2]

MARK SCHEME

7(d)(i)	1 mark per bullet point to max 2 • Prevents unauthorised access to the data • Monitors incoming and outgoing traffic • Blocks transmissions from unauthorised sources / websites / ports • Maintains an allow list / deny list of IP addresses	2
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9608_s20_qp_12_sc_7.d.ii

QUESTION

- (ii) Give **one additional** security measure that the company can use to protect the data on the servers from online threats.

..... [1]

MARK SCHEME

7(d)(ii)	1 mark only from: - Running up-to-date antivirus software - Use of proxy server - Strong / Biometric passwords - Etc.	1
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9608_s20_qp_11_sc_1.b

QUESTION

- (b) The program is installed on a computer system that has security measures in place to protect its data.

Complete the following table.

Security measure	Description
.....	Data are written on two or more disks simultaneously.
Encryption	
.....	A copy of the data is taken and stored in another location.

[3]

MARK SCHEME

1(b)	1 mark for each correct entry		3
	Security measure	Description	
	Disk mirroring	Data are written on two or more disks simultaneously.	
	Encryption	Contents are scrambled so they cannot be understood without a decryption key	
	Backup	A copy of the data is taken and stored in another location	

9608_s19_qp_13_sc_2.b

QUESTION

(b) Computer systems can be protected by physical methods such as locks.
Describe **two** non-physical methods used to improve the security of computer systems.

1

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2

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[6]

MARK SCHEME

2(b)	1 mark for identifying method 2 marks for description to max 2 × 3 For example: User accounts User has a username and password Access to resources can be limited to specific accounts Cannot access <u>system</u> without valid username and password // prevents unauthorised access to the <u>system</u> Firewall All incoming and outgoing network traffic goes through firewall Blocks signals that do not meet requirements Keeps a log of signals Applications can have network access restricted Anti-malware Scans for malicious software Quarantines or deletes any malicious software found Scans can be scheduled at regular intervals Should be kept up to date Auditing Logging all actions/changes to the system In order to identify any unauthorised use Application Security (accept equivalent) Applying regular updates / patches Finding, fixing and preventing security vulnerabilities in any (installed) application	6
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9608_w18_qp_13_sc_5.b

QUESTION

- (b) Users need to enter their name and email address to create an account. The information is stored in a database on Arnold's computer.

Give **three** ways that Arnold can ensure users' details are kept secure.

- 1
- 2
- 3

[3]

MARK SCHEME

5(b)	1 mark per bullet point, max 3 • Firewall / proxy • Encryption • Username and Password • Physical security • Biometric authentication // by example • Two-step authentication // by example • Anti-malware	3
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9608_w18_qp_12_sc_6.a

QUESTION

- 6** Kim is using her laptop computer to write a program in a high-level language.
- (a)** Kim needs to make sure the program is secure against unauthorised access. She has already set up a username and password on her laptop.
- Identify **two** additional electronic measures that Kim can use to keep the program secure.
- 1
- 2 [2]

MARK SCHEME

6(a)	1 mark for each method to max 2 • Biometric authentication // by example • Two-step authentication // by example • Firewall / proxy • Encryption • Different access rights for different users • Password protect <u>file</u> (using a different password) • Anti-malware	2
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9608_w17_qp_13_sc_3.a

QUESTION

- 3 A Local Area Network is used by staff in a hospital to access data stored in a Database Management System (DBMS).

(a) Name **two** security measures to protect computer systems.

1

2 [2]

MARK SCHEME

3(a)	Two marks from: Physical measures Access rights Encryption Firewall Use authentication methods such as usernames and passwords Anti-malware program	Max 2
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9608_s17_qp_13_sc_6.b.iv

QUESTION

- (iv) There have been some unexplained images and advertisements appearing on the screen. The user suspects it is malware.

Utility software solution [1]

MARK SCHEME

6(b)(iv)	Anti-virus software	1
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9608_s17_qp_11_sc_6.b.iv

QUESTION

- (iv) There have been some unexplained images and advertisements appearing on the screen. The user suspects it is malware.

Utility software solution [1]

MARK SCHEME

6(b)(iv)	Anti-virus software	1
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9608_w16_qp_13_sc_7.c.iii

QUESTION

- (iii) *"The use of passwords will always prevent unauthorised access to the data stored on the computers."*

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.....
.....
.....[2]

MARK SCHEME

- (iii) • A password does not prevent unauthorised access, it makes it more difficult
• Password can be guessed (if weak) // Password can be stolen // A relevant example of misappropriation of password [2]

9608_w16_qp_11_sc_7.c.i

QUESTION

- (c) Employees using the new computers receive training. At the end of the training, each employee completes a series of questions.

Three answers given by an employee are shown below.

Explain why each answer is incorrect.

- (i) *"Encryption prevents hackers breaking into the company's computers."*

.....
.....
.....
.....[2]

MARK SCHEME

- (c) (i) • Hackers can still access the data (and corrupt it, change it or delete it)
• Encryption simply makes data incomprehensible (without decryption key / algorithm) [2]

QUESTION

- (iii) *"The use of passwords will always prevent unauthorised access to the data stored on the computers."*

.....

[2]

MARK SCHEME

- (iii) • A password does not prevent unauthorised access, it makes it more difficult
 • Password can be guessed (if weak) // Password can be stolen // A relevant example of misappropriation of password [2]

QUESTION

- (b) Describe **three** security measures that the bank could implement to protect its electronic data.

Security measure 1
 Description

 Security measure 2
 Description

 Security measure 3
 Description

[6]

MARK SCHEME

(b) Three pairs from:

[6]

- Installing a firewall and ensuring it is switched on.
- To stop unauthorised access / hackers gaining access to the bank's computer network.
- Use authentication methods such as passwords and usernames.
- Passwords should be strong / biometrics.
- Encrypt the data.
- So that if data is accessed it will be meaningless / only accessed by those with decryption key.
- Set up access rights...
- To stop users reading/editing data they are not permitted to access.
- Installing and running an up to date anti-malware program (anti-virus/anti-spyware etc.).
- To detect / remove / quarantine viruses / key-loggers etc.
- Make regular backups of the data.
- To separate device or off site to enable recovery if necessary.
- Employ measures for physical security.
- Example of a measure for physical security.

9608_w15_qp_13_sc_10.b

QUESTION

(b) A virus checker has been installed on a PC.

Give **two** examples of when a virus checker should perform a check.

1

.....

2

.....

[2]

MARK SCHEME

(b) any **two** from:

- checks for boot sector viruses when machine is first turned on
- when an external storage device is connected
- checks a file / web page when it is accessed / downloaded

[2]

QUESTION

(b) A virus checker has been installed on a PC.

Give **two** examples of when a virus checker should perform a check.

1

.....

2

.....

[2]

MARK SCHEME

(b) any **two** from:

- checks for boot sector viruses when machine is first turned on
- when an external storage device is connected
- checks a file / web page when it is accessed / downloaded

[2]

QUESTION

3 (a) Give the definition of the terms firewall and authentication. Explain how they can help with the security of data.

Firewall

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.....

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Authentication

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.....

.....

[3]

MARK SCHEME

- 3 (a)** maximum of **two marks** for firewall description + maximum of **two marks** for authentication description

Firewall

- sits between the computer or LAN and the Internet/WAN and permits or blocks traffic to/from the network
- can be software and/or hardware
- software firewall can make precise decisions about what to allow or block as it can detect illegal attempts by specific software to connect to Internet
- can help to block hacking or viruses reaching a computer

Authentication

- process of determining whether somebody/something is who/what they claim to be
- frequently done through log on passwords/biometrics
- because passwords can be stolen/cracked, digital certification is used
- helps to prevent unauthorised access to data

[3]

9608_s15_qp_12_sc_6.b.ii

QUESTION

- (ii) Describe **one** way of ensuring the security of the data against malicious damage.

.....
.....[1]

MARK SCHEME

- (ii) Any **one** from:

- protection of data (or equivalent) with passwords/using password and username for logging on include e.g. fingerprint scanning
- encryption
- installation and use of up to date anti-malware/anti-virus
- give different access rights to different users
- use a firewall,
- physical methods/lock doors and use secure entry devices/CCTV

[1]

QUESTION

(ii) Describe **one** way of ensuring the security of the data against malicious damage.

.....
.....[1]

MARK SCHEME

(ii) Any **one** from:

- protection of data (or equivalent) with passwords/using password and username for logging on include e.g. fingerprint scanning
- encryption
- installation and use of up to date anti-malware/anti-virus
- give different access rights to different users
- use a firewall,
- physical methods/lock doors and use secure entry devices/CCTV

[1]

Show understanding of the threats to computer and data security posed by networks and the internet

9608_w15_qp_13_sc_10.a

QUESTION

10 (a) Explain the term computer virus.

.....

.....

.....

.....[2]

MARK SCHEME

10 (a) any **two** from:

- malicious code / software / program
- that replicates / copies itself
- can cause loss of data / corruption of data on the computer
- can cause computer to “crash” / run slowly
- can fill up hard disk with data

[2]

9608_w15_qp_11_sc_10.a

QUESTION

10 (a) Explain the term computer virus.

.....

.....

.....

.....[2]

MARK SCHEME

10 (a) any **two** from:

- malicious code / software / program
- that replicates / copies itself
- can cause loss of data / corruption of data on the computer
- can cause computer to “crash” / run slowly
- can fill up hard disk with data

[2]

Describe methods that can be used to restrict the risks posed by threats

9608_s20_qp_12_sc_7.d.ii

QUESTION

(ii) Give **one additional** security measure that the company can use to protect the data on the servers from online threats.

..... [1]

MARK SCHEME

7(d)(ii)	1 mark only from: • Running up-to-date antivirus software • Use of proxy server • Strong / Biometric passwords • Etc.	1
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9608_s15_qp_11_sc_6.b.ii

QUESTION

(ii) Describe **one** way of ensuring the security of the data against malicious damage.

.....

.....[1]

MARK SCHEME

(ii) Any one from: • protection of data (or equivalent) with passwords/using password and username for logging on include e.g. fingerprint scanning • encryption • installation and use of up to date anti-malware/anti-virus • give different access rights to different users • use a firewall, • physical methods/lock doors and use secure entry devices/CCTV	[1]
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Describe security methods designed to protect the security of data

9608_s21_qp_11_sc_2.d

QUESTION

(d) Zak's company holds details about clients in a database.

Give **three** security measures that Zak can implement to make sure that only authorised employees can access the data.

1

2

3 [3]

MARK SCHEME

2(d)	1 mark per bullet point to max 3 • Password • Biometrics • Access rights • Swipe cards • Physical access measures, e.g. security guards • Implement a firewall/proxy to monitor remote access requests • Two-step authentication	3
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QUESTION

5 Wei is developing a program.

(a) He wants to make sure the source code is secure on his laptop.

Explain how encrypting the source code can keep it secure.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

5(a)	1 mark per bullet point to max 3 • Encryption scrambles the source code (so it is meaningless) • ... using an encryption key / algorithm • If the file is accessed without authorisation it will be meaningless • It requires a decryption key / algorithm to unscramble	3
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QUESTION

- (b) The program is installed on a computer system that has security measures in place to protect its data.

Complete the following table.

Security measure	Description
.....	Data are written on two or more disks simultaneously.
Encryption	
.....	A copy of the data is taken and stored in another location.

[3]

MARK SCHEME

1(b)	1 mark for each correct entry	3								
	<table><tr><th>Security measure</th><th>Description</th></tr><tr><td>Disk mirroring</td><td>Data are written on two or more disks simultaneously.</td></tr><tr><td>Encryption</td><td>Contents are scrambled so they cannot be understood without a decryption key</td></tr><tr><td>Backup</td><td>A copy of the data is taken and stored in another location</td></tr></table>	Security measure	Description	Disk mirroring	Data are written on two or more disks simultaneously.	Encryption	Contents are scrambled so they cannot be understood without a decryption key	Backup	A copy of the data is taken and stored in another location	
Security measure	Description									
Disk mirroring	Data are written on two or more disks simultaneously.									
Encryption	Contents are scrambled so they cannot be understood without a decryption key									
Backup	A copy of the data is taken and stored in another location									

QUESTION

- (ii) The supermarket is concerned about the security and integrity of the data on the server.

Identify **two** methods that can be used to minimise the security risk to the data, and **one** method to protect the integrity of the data.

Security 1

.....

Security 2

.....

Integrity

.....

[3]

MARK SCHEME

1(c)(ii)	1 mark for each security method to max 2, 1 mark for integrity Security - encryption - access rights - username and password // biometrics // user accounts - backup // disk mirroring - firewall - Physical methods (e.g. CCTV, locked rooms etc.) Integrity - checksum - parity - validation on input	3
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QUESTION

- (b) The software development company uses data backup and disk-mirroring to keep their data secure.

Explain how data backup and disk-mirroring allow the company to recover from data loss.

Data backup

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.....

.....

Disk-mirroring

.....

.....

.....

[4]

MARK SCHEME

4(b)	1 mark per bullet point to max 2 · 2 Data backup A copy of data will have been made and stored elsewhere. If the original is lost, the backup can be used to restore the data. Disk-mirroring The data is stored on two disks simultaneously. If the first disk drive fails, the data is accessed from the second disk.	4
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QUESTION

- (b) Users need to enter their name and email address to create an account. The information is stored in a database on Arnold's computer.

Give **three** ways that Arnold can ensure users' details are kept secure.

1

2

3

[3]

MARK SCHEME

5(b)	1 mark per bullet point, max 3 • Firewall / proxy • Encryption • Username and Password • Physical security • Biometric authentication // by example • Two-step authentication // by example • Anti-malware	3
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9608_w18_qp_12_sc_6.a

QUESTION

6 Kim is using her laptop computer to write a program in a high-level language. (a) Kim needs to make sure the program is secure against unauthorised access. She has already set up a username and password on her laptop. Identify two additional electronic measures that Kim can use to keep the program secure. 1 2 [2]
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MARK SCHEME

6(a)	1 mark for each method to max 2 • Biometric authentication // by example • Two-step authentication // by example • Firewall / proxy • Encryption • Different access rights for different users • Password protect <u>file</u> (using a different password) • Anti-malware	2
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QUESTION

(b) Name and describe **two** security measures the Network Manager has in place to protect the security of the data held in the DBMS.

1

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2

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[4]

MARK SCHEME

3(b)	1 Mark for stating the security measure 1 Mark for a corresponding description Maximum 2 marks for each measure Maximum 2 measures Physical measures Locked doors/keyboards etc. Secure methods of access, keypads/ biometric scans etc. Backup of data Regular copies of the data are made If the data is corrupted it can be restored Disk-mirroring All activity is duplicated to a second disk in real time so that if the first disk fails there is a complete copy available Access rights Different access rights for individuals/groups of users To stop users editing data they are not permitted to access By example Encryption If accessed, data cannot be understood by unauthorised personnel Accessed only by those with the decryption key Firewall To stop unauthorised access/hackers gaining access to the computer network Use authentication methods such as passwords and usernames Passwords should be strong / biometrics To prevent unauthorised access to data Anti-malware program To detect / remove / quarantine viruses / key-loggers etc. Carrying out regular scans Concurrent Access Controls // Record locking Closes a record to second user until first update complete To prevent simultaneous updates being lost	4
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QUESTION

- (c) Employees using the new computers receive training. At the end of the training, each employee completes a series of questions.

Three answers given by an employee are shown below.

Explain why each answer is incorrect.

- (i) *"Encryption prevents hackers breaking into the company's computers."*

.....

[2]

MARK SCHEME

- (c) (i) • Hackers can still access the data (and corrupt it, change it or delete it)
 • Encryption simply makes data incomprehensible (without decryption key / algorithm)
 [2]

QUESTION

- (c) Employees using the new computers receive training. At the end of the training, each employee completes a series of questions.

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Explain why each answer is incorrect.

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.....

[2]

MARK SCHEME

- (c) (i) • Hackers can still access the data (and corrupt it, change it or delete it)
 • Encryption simply makes data incomprehensible (without decryption key / algorithm)
 [2]

QUESTION

(b) Describe **three** security measures that the bank could implement to protect its electronic data.

Security measure 1

Description

.....

.....

Security measure 2

Description

.....

.....

Security measure 3

Description

.....

.....[6]

MARK SCHEME

(b) Three pairs from:

[6]

- Installing a firewall and ensuring it is switched on.
- To stop unauthorised access / hackers gaining access to the bank's computer network.
- Use authentication methods such as passwords and usernames.
- Passwords should be strong / biometrics.
- Encrypt the data.
- So that if data is accessed it will be meaningless / only accessed by those with decryption key.
- Set up access rights...
- To stop users reading/editing data they are not permitted to access.
- Installing and running an up to date anti-malware program (anti-virus/anti-spyware etc.).
- To detect / remove / quarantine viruses / key-loggers etc.
- Make regular backups of the data.
- To separate device or off site to enable recovery if necessary.
- Employ measures for physical security.
- Example of a measure for physical security.

QUESTION

- 3 (a) Give the definition of the terms firewall and authentication. Explain how they can help with the security of data.

Firewall

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.....

.....

.....

.....

Authentication

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.....

.....

[3]

MARK SCHEME

- 3 (a) maximum of **two marks** for firewall description + maximum of **two marks** for authentication description

Firewall

- sits between the computer or LAN and the Internet/WAN and permits or blocks traffic to/from the network
- can be software and/or hardware
- software firewall can make precise decisions about what to allow or block as it can detect illegal attempts by specific software to connect to Internet
- can help to block hacking or viruses reaching a computer

Authentication

- process of determining whether somebody/something is who/what they claim to be
- frequently done through log on passwords/biometrics
- because passwords can be stolen/cracked, digital certification is used
- helps to prevent unauthorised access to data

[3]

QUESTION

(b) All of the uses in **part (a)** involve the input or output of data.

(i) Describe **two** methods of preventing accidental loss of data.

1

.....

2

.....[2]

MARK SCHEME

(b) (i) Any **two** from:

- frequent (or equivalent) backup EITHER to secondary media/to 3rd party server/cloud/removable devices/continuous backup OR stored remotely
- disk-mirroring strategy/RAID
- UPS (uninterruptable power supply)/backup generator

[2]

QUESTION

(ii) Describe **one** way of ensuring the security of the data against malicious damage.

.....

.....[1]

MARK SCHEME

(ii) Any **one** from:

- protection of data (or equivalent) with passwords/using password and username for logging on include e.g. fingerprint scanning
- encryption
- installation and use of up to date anti-malware/anti-virus
- give different access rights to different users
- use a firewall,
- physical methods/lock doors and use secure entry devices/CCTV

[1]

QUESTION

(b) All of the uses in **part (a)** involve the input or output of data.

(i) Describe **two** methods of preventing accidental loss of data.

- 1
-
- 2
-[2]

MARK SCHEME

(b) (i) Any **two** from:

- frequent (or equivalent) backup EITHER to secondary media/to 3rd party server/cloud/removable devices/continuous backup OR stored remotely
- disk-mirroring strategy/RAID
- UPS (uninterruptable power supply)/backup generator

[2]